Graphical Abstract Caption

Caption:
This graphical abstract illustrates the Geodesic Diagnostic Trajectories (GDT) framework for clinical EHR retrieval. **Left:** Standard retrieval systems suffer from "stale context pollution" when clinicians make abrupt diagnostic query pivots during a chart review (e.g., shifting abruptly from sepsis markers to renal dosing). **Center:** GDT solves this by modeling search intents on a Riemannian manifold ($\mathcal{S}_d^+$). It utilizes a Geometric Subspace Interference (GSI) gate (κ_t) to explicitly suppress outdated context, and a Joint Embedding Predictive Architecture (JEPA) for predictive prefetching. **Right:** Empirical results demonstrate that GDT reduces retrieval latency by 26.9% (achieving 138 ms) compared to traditional TF-IDF and BM25 baselines. Furthermore, the framework successfully verifies formal interference bounds (Proposition 1) across 100% of simulated transitions, translating to an estimated 4.1 minutes saved per clinical reviewer per 4-hour shift.